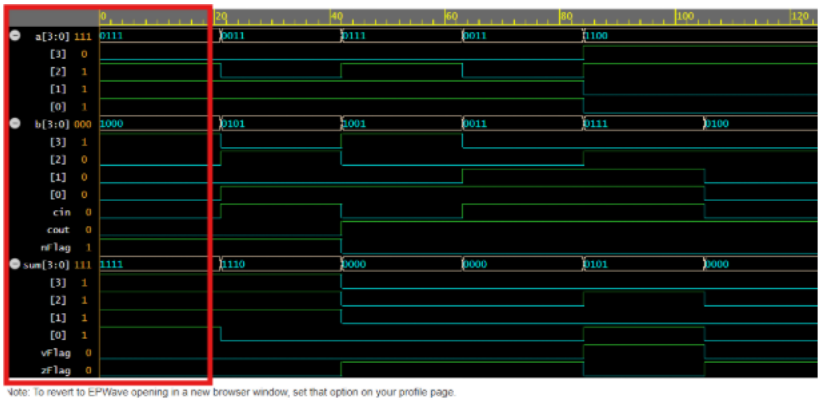
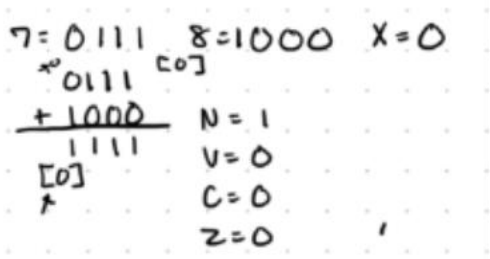
Timing Diagrams:

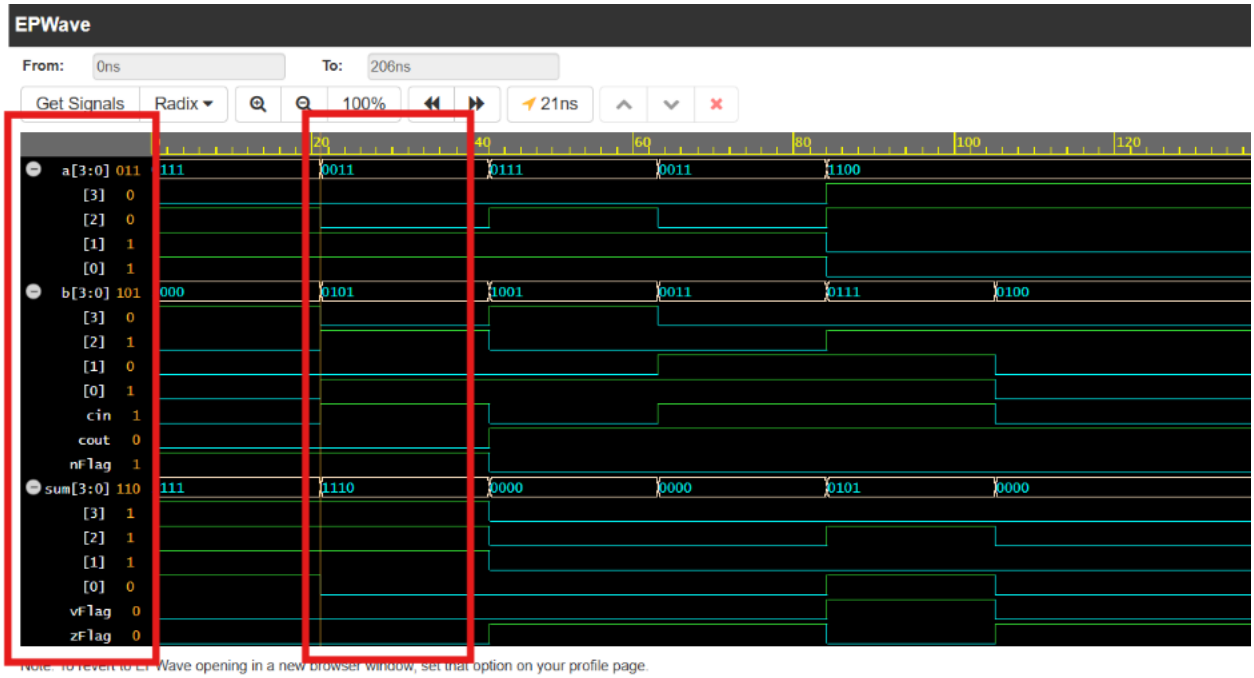


Predicted Answer:

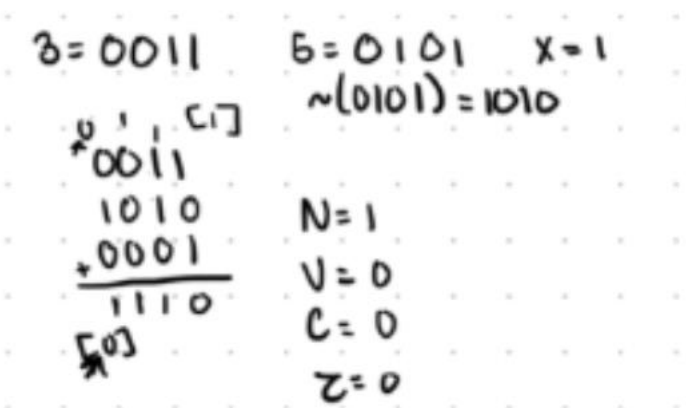


With the A = 7 = 0111, B = 8 = 1000, and X = cin = 0 the full adder would be performing addition. After adding (performing the OR function), the sum is 15 = 1111. Since the MSB = 1, there would be an N flag. The carry out value is 0 which is C. And since it's not a string of zeros, the Z flag wouldn't go off. By XOR the c3in and c3out, it would be 1 XOR 1 = 0 causing the V flag not to go out. Based off of the time diagram at #0, the sum [3:0] matches 1111, and used #1 time units for displaying the NVCZ flags.

Time Diagram:

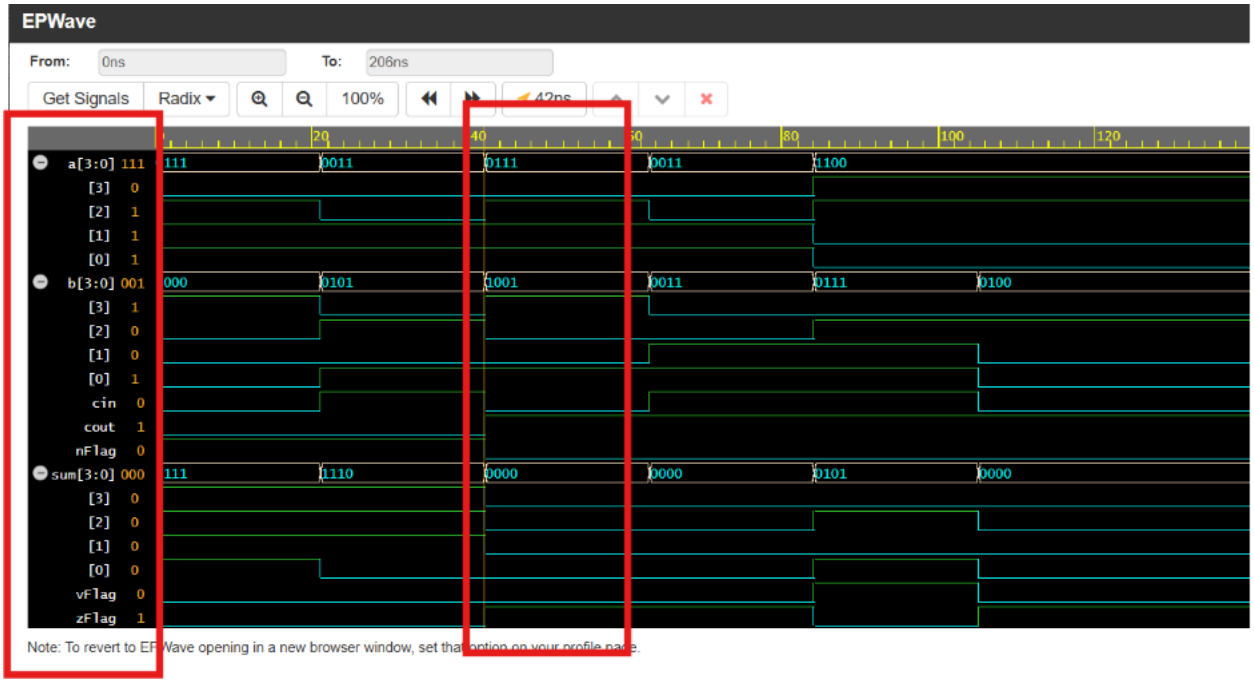


Predicted Answer:

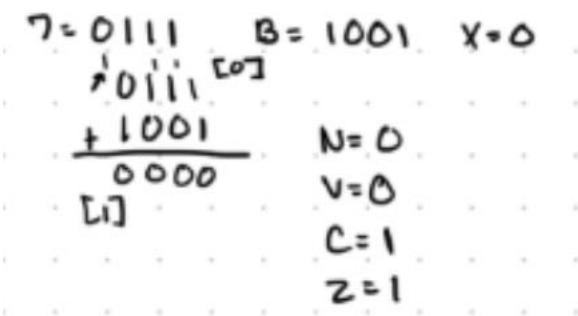


With the A = 3 = 0011, B = 5 = 0101, and X = cin = 1 the full adder would be performing subtraction. With performing subtraction, to get the accurate result we must perform addition with the A value, the not of B value, and 0001 = 1 for the sum. The sum with this operation is 14 = 1110. Since the MSB = 1, there would be an N flag. The carry out value is 0 which is C. And since it's not a string of zeros, the Z flag wouldn't go off. Based from the time diagram at #20, the sum [3:0] matches 1110, and used #1 time units for displaying the NVCZ flags.

Time Diagram:

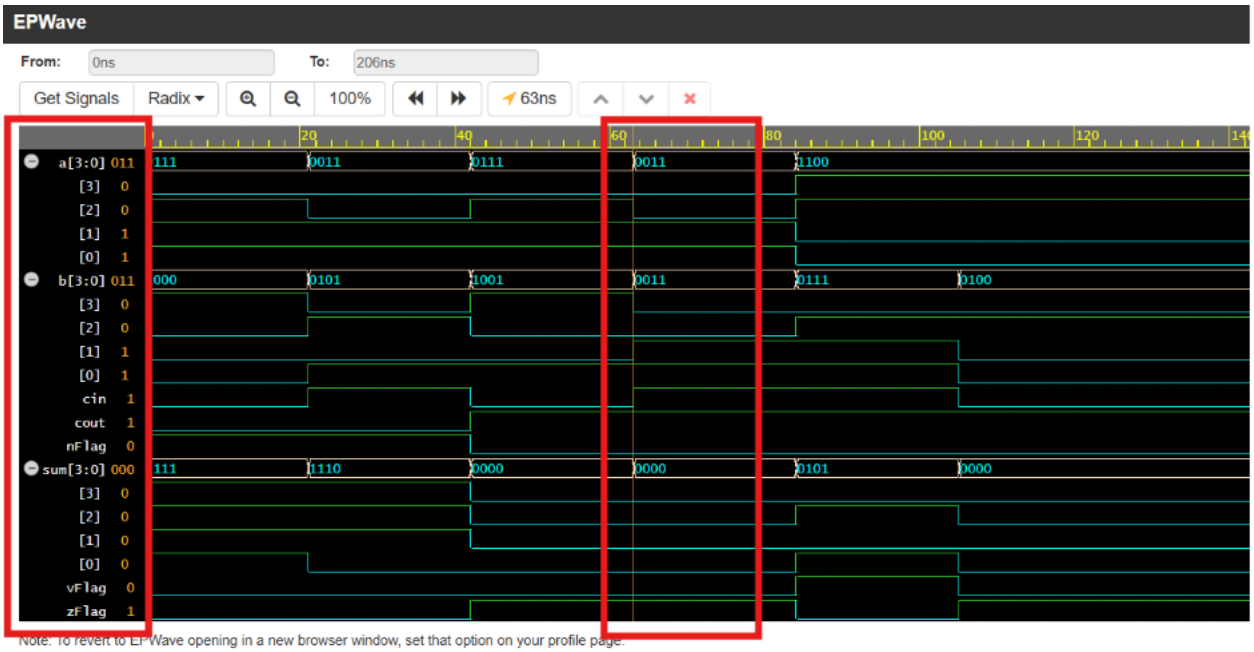


Predicted Answer:

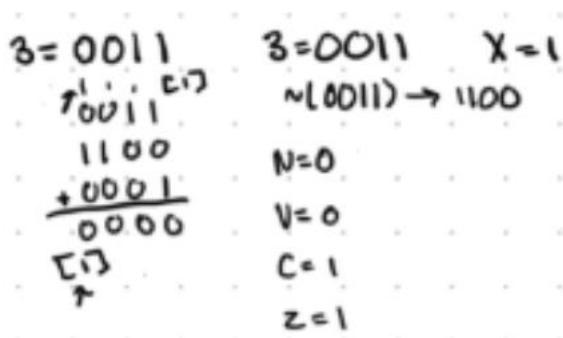


With the A = 7 = 0111, B = 5 = 1001, and X = cin = 0 the full adder would be performing addition. After adding (performing the OR function), the sum is 0 = 0000. Since the MSB = 0, there wouldn't be an N flag. The carry out value is 1 which is C. And since there is a string of zeros, the Z flag would go off. By XOR the c3in and c3out, it would be 1 XOR 1 = 0 causing the V flag not to go out. Based off of the time diagram at #40, the sum [3:0] matches 0000, and used #1 time units for displaying the NVCZ flags.

Time Diagram:

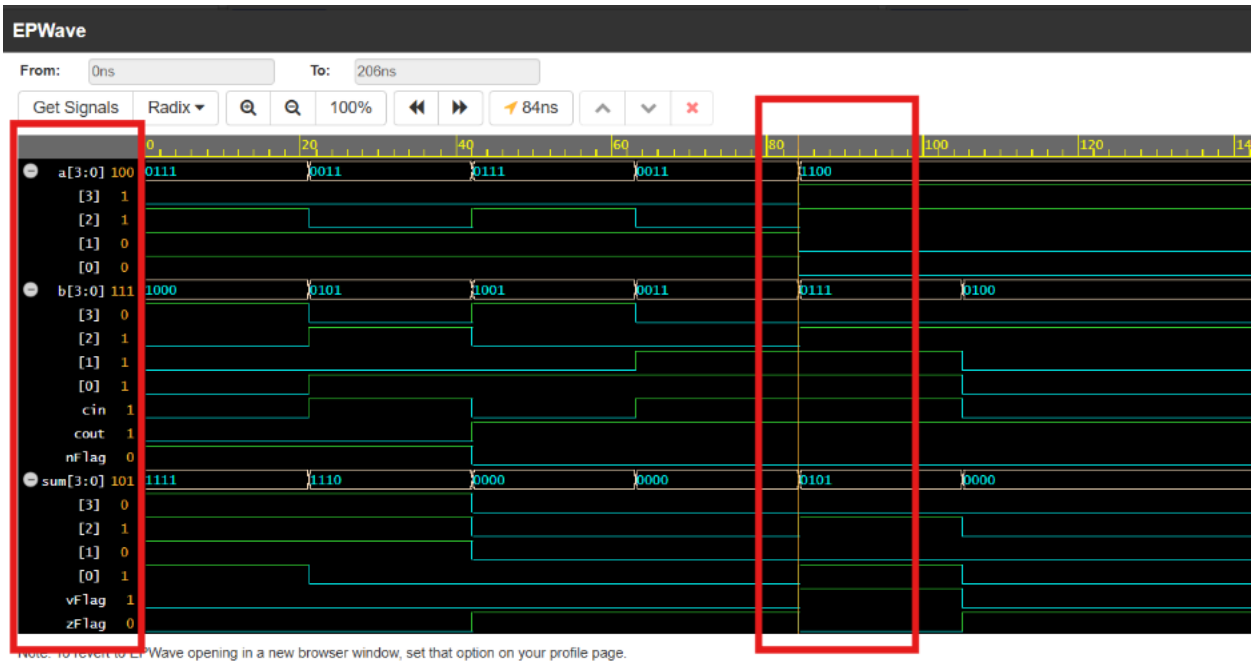


Predicted Answer:

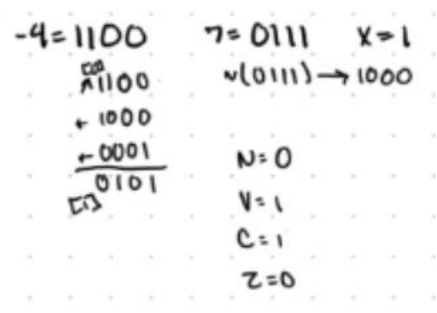


With the A = 3 = 0011, B = 5 = 0011, and X = cin = 1 the full adder would be performing subtraction. With performing subtraction, to get the accurate result we must perform addition with the A value, the not of B value, and 0001 = 1 for the sum. The sum with this operation is 0 = 0000. Since the MSB = 0, there wouldn't be an N flag. The carry out value is 1 which is C. And since there is a string of zeros, the Z flag would go off. By XOR the c3in and c3out, it would be 1 XOR 1 = 0 causing the V flag not to go out. Based from the time diagram at #60, the sum [3:0] matches 0000, and used #1 time units for displaying the NVCZ flags.

Timed Diagram:

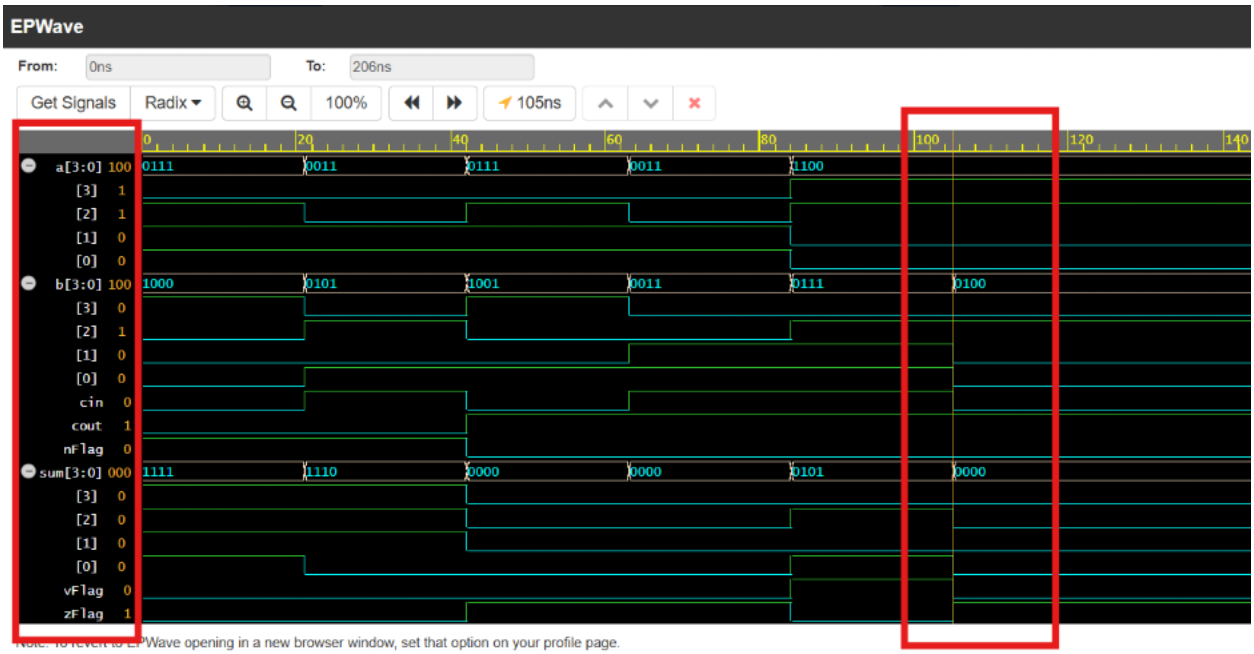


Predicted Answer:

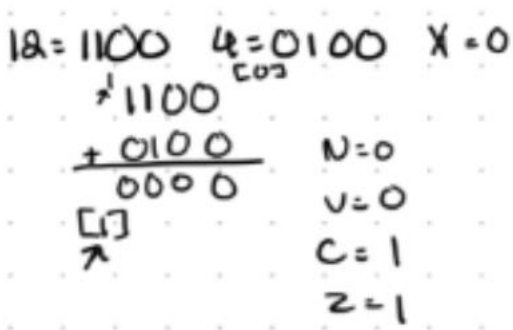


With the A = 3 = 1100, B = 5 = 0111, and X = cin = 1 the full adder would be performing subtraction. With performing subtraction, to get the accurate result we must perform addition with the A value, the not of B value, and 0001 = 1 for the sum. The sum with this operation is 5 = 0101. Since the MSB = 0, there wouldn't be an N flag. The carry out value is 1 which is C. And since it's not a string of zeros, the Z flag wouldn't go off. By XOR the c3in and c3out, it would be 0 XOR 1 = 1 causing the V flag to turn on. Based from the time diagram at #80, the sum [3:0] matches 0101, and used #1 time units for displaying the NVCZ flags.

Timed Diagram:



Predicted Answer:



With the A = 7 = 1100, B = 5 = 0100, and X = cin = 0 the full adder would be performing addition. After adding (performing the OR function), the sum is 0 = 0000. Since the MSB = 0, there wouldn't be an N flag. The carry out value is 1 which is C. And since there is a string of zeros, the Z flag would go off. By XOR the c3in and c3out, it would be 1 XOR 1 = 0 causing the V flag not to go out. Based off of the time diagram at #100, the sum [3:0] matches 0000, and used #1 time units for displaying the NVCZ flags.